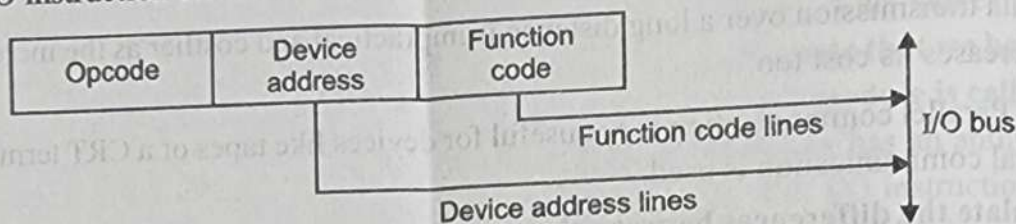


7.7 I/O AND MEMORY BUSES

CPU issues I/O command simultaneously, with address of device to which it is meant. The command decoder in interface module decodes and specified command is activated in interface logic. The I/O instruction format is as follows:



It consists of **opcode** which specifies that instruction an I/O instruction device address, which specify to which device, command is meant for and **function code** specifying command for I/O interface.

Types of I/O commands:

There are 4 types of I/O commands that an interface may receive when addressed by CPU. They are as follows:

1. Control Command

This command issued to activate peripherals.

E.g. Rewind tape for tape drives.

2. Test Command

This is issued to interface by CPU to test the conditions of I/O device and conditions in interface, such as buffer full, device not ready etc.

3. Data Output Command

(a) Issued by CPU to interface when it outputs data to peripheral.

(b) Interface accepts data from I/O bus and stores in buffer register for peripheral.

4. Data Input Command

(a) Issued by CPU when it wants to input data from peripheral.

(b) Interface receives data from peripheral and stores in buffer register for CPU.

7.8 I/O INTERFACING TECHNIQUES

Many processors (microprocessors) isolate memory and I/O system. It partitions memory from I/O using software's. This software has instructions that specifically address memory and other instruction that specifically access I/O. **Please understand that when these instructions are decoded by the processor, an appropriate control signal is generated to activate either memory or I/O operation.** So I/O devices can be interfaced to a computer system's I/O in two ways-

(a) Memory mapped I/O.

(b) I/O mapped I/O.

In 8085, for instance, this is done by the M/IO signal pin on its IC.

Note: In 8085,

- (a) $M/I/O' = 1$ means processor is communicating with memory.
 (b) $M/I/O' = 0$ means processor is communicating with I/O.
 Let us study these two techniques now.

I. Memory – mapped I/O

A method in which the total memory address space is partitioned is shown in Fig. 7.9

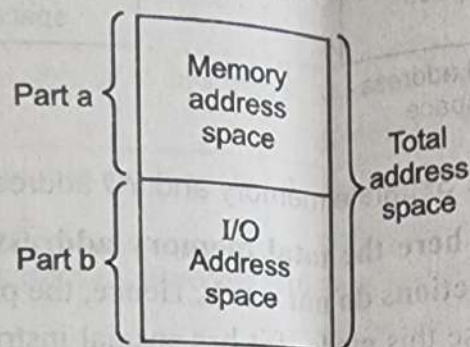


Fig. 7.9. Division of total address space

In this technique, a memory reference instruction that causes data to be fetched from or stored at the address specified, automatically becomes an I/O instruction if that address is made an address of an I/O port. The usual memory related instructions are used for I/O related operations. The special I/O instructions are not required. **Please remember the following points regarding memory mapped I/O:-**

1. I/O devices, here, are treated as memory locations.
2. There will be no separation like memory or I/O.
3. Each I/O device will have 16 bit address.
4. The control signals used will be the same as memory ($MEMR'$ and $MEMW'$).
5. Interfacing between the I/O and processor will be same as single memory location.
6. The number of device I/O will be 64 K, (shared by I/O and memory).

We now draw, Fig. 7.10 that show programmed I/O with memory mapped I/O.

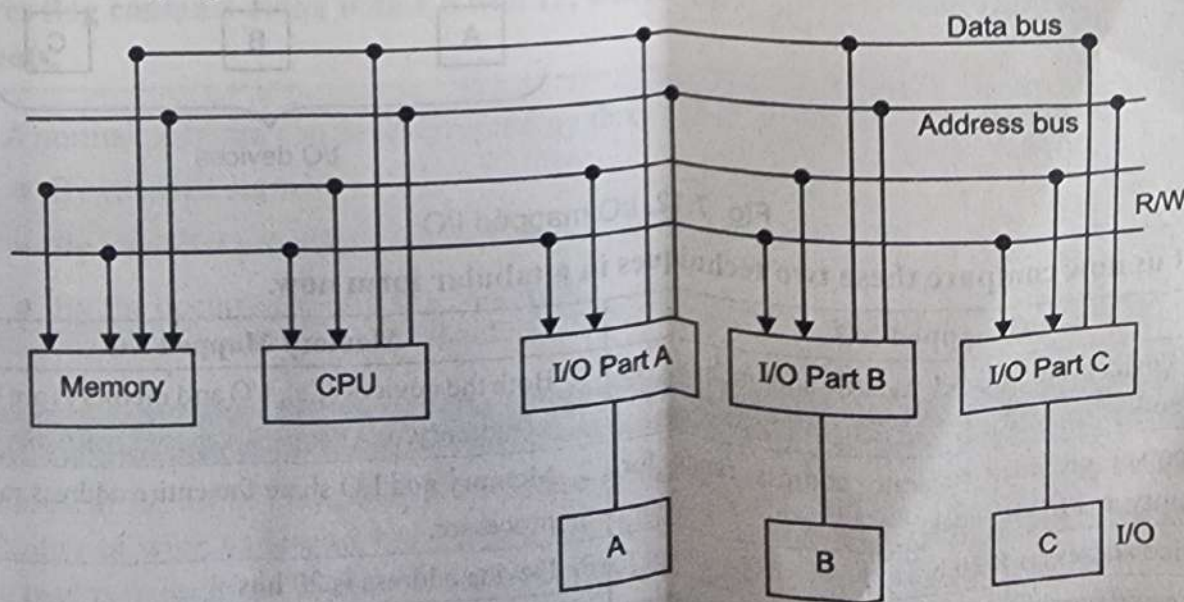


Fig 7.10

II. I/O Mapped I/O

When we divide the total address space as in method-I we are actually reducing memory address space. To avoid this reduction, we allocate a different I/O address space. This is shown in Fig 7.11.

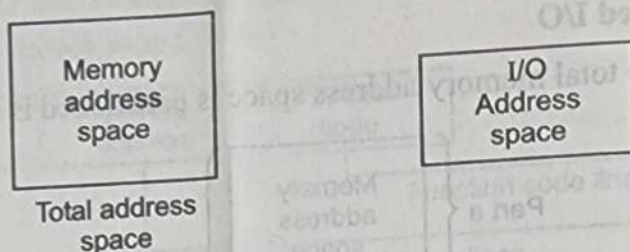


Fig 7.11. Separate memory and I/O address space

Please note that although here the total memory address space has not been used but now the memory related instructions do not work. Hence, the processor can only use this mode. Hence, the processor can only use this mode if it has special instructions for I/O related operation (like I/O read, I/O write etc). Also note the following points regarding I/O Mapped I/O:-

1. Here, the I/O device is treated as I/O device only.

2. Each I/O device uses 8 bits of address lines.

Hence, $2^8 = 256$ I/O devices can be connected to the system.

3. 256 input and 256 output devices can be connected. We now draw Fig. 7.12 that shows programmed I/O with I/O mapped I/O.

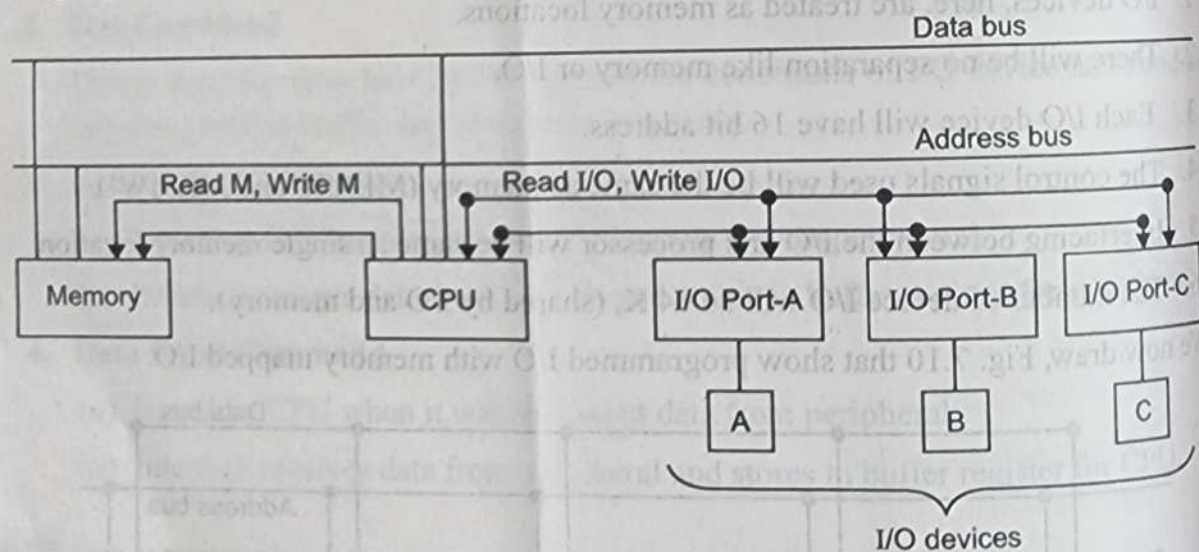


Fig. 7.12. I/O mapped I/O

Let us now compare these two techniques in a tabular form now.

I/O mapped I/O	Memory Mapped I/O
1. I/O devices are treated as I/O and memory as memory.	1. Both the devices (i.e., I/O and memory) are treated as memory.
2. Processor provides separate address range for memory and I/O devices.	2. Memory and I/O share the entire address range of processor.
3. Device address is 8/16 bits.	3. Device address is 20 bits.
4. Processor provides less address lines for accessing I/O hence, less decoding is required.	4. Processor provides more address lines for accessing memory. So, more decoding is required.

I/O mapped I/O

5. Control signals used for I/O are IOR' and IOW'.
6. Special instructions like IN and OUT are used.
7. Data transfers between accumulator and I/O devices is allowed.
8. Lesser hardware is required as only 8/16 bits of address decoding is required.
9. Arithmetic and logical operations are not possible with direct data from I/O devices.
10. Total number of devices that can be connected are –
1 MB memory.
65,535 (max) input devices.
65,535 (max) output devices.

Memory Mapped I/O

5. Control signals used for I/O are MEMR' and MEMW'.
6. All memory related instruction can be used to communicate with I/O.
7. Data transfer is possible between any register and I/O.
8. More hardware is required as 20 bits address decoding is needed.
9. Arithmetic and logical operations can be performed on direct data from I/O device.
10. Total number of devices that can be connected will be only 1 MB as it is shared between memory and I/O.